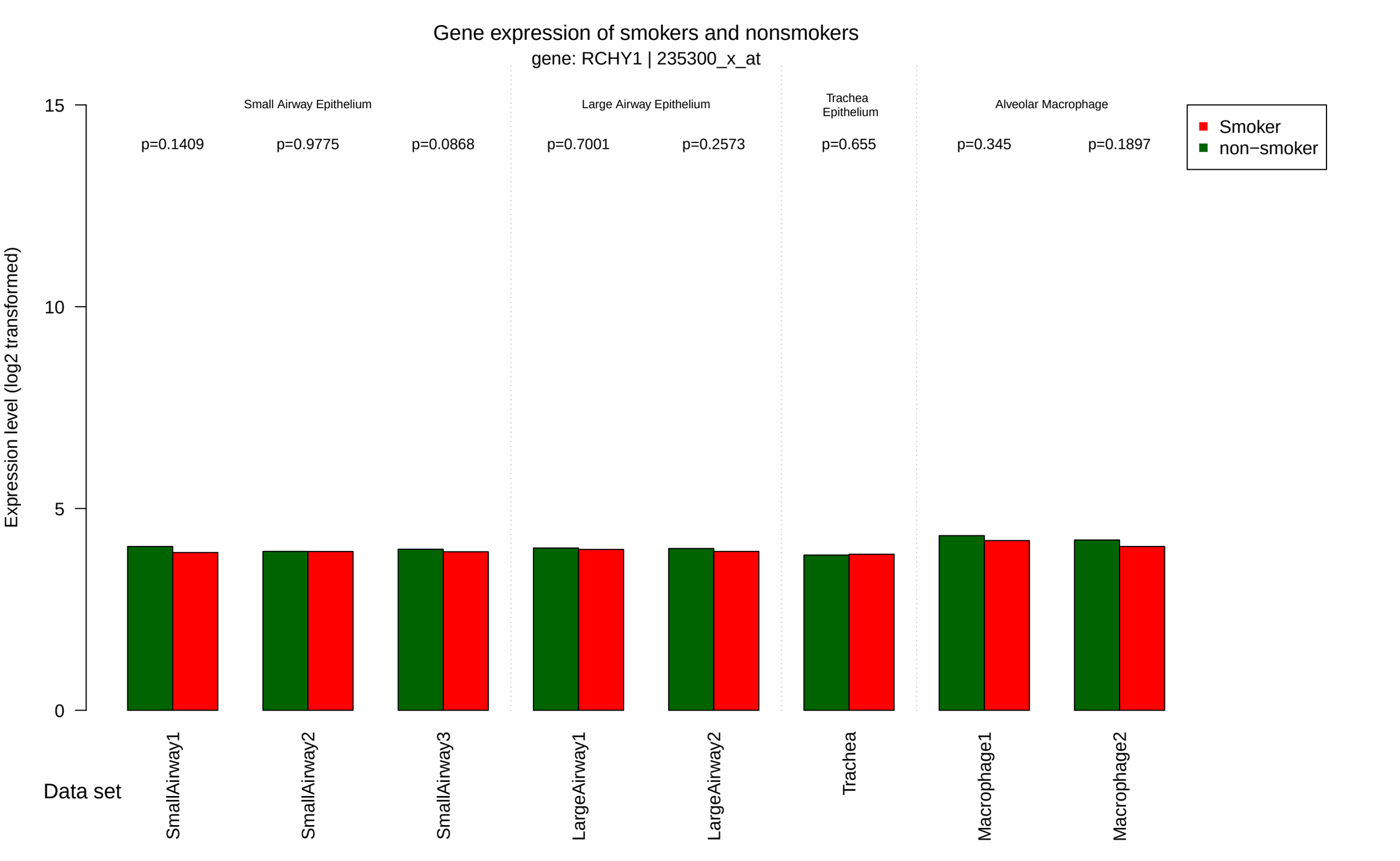
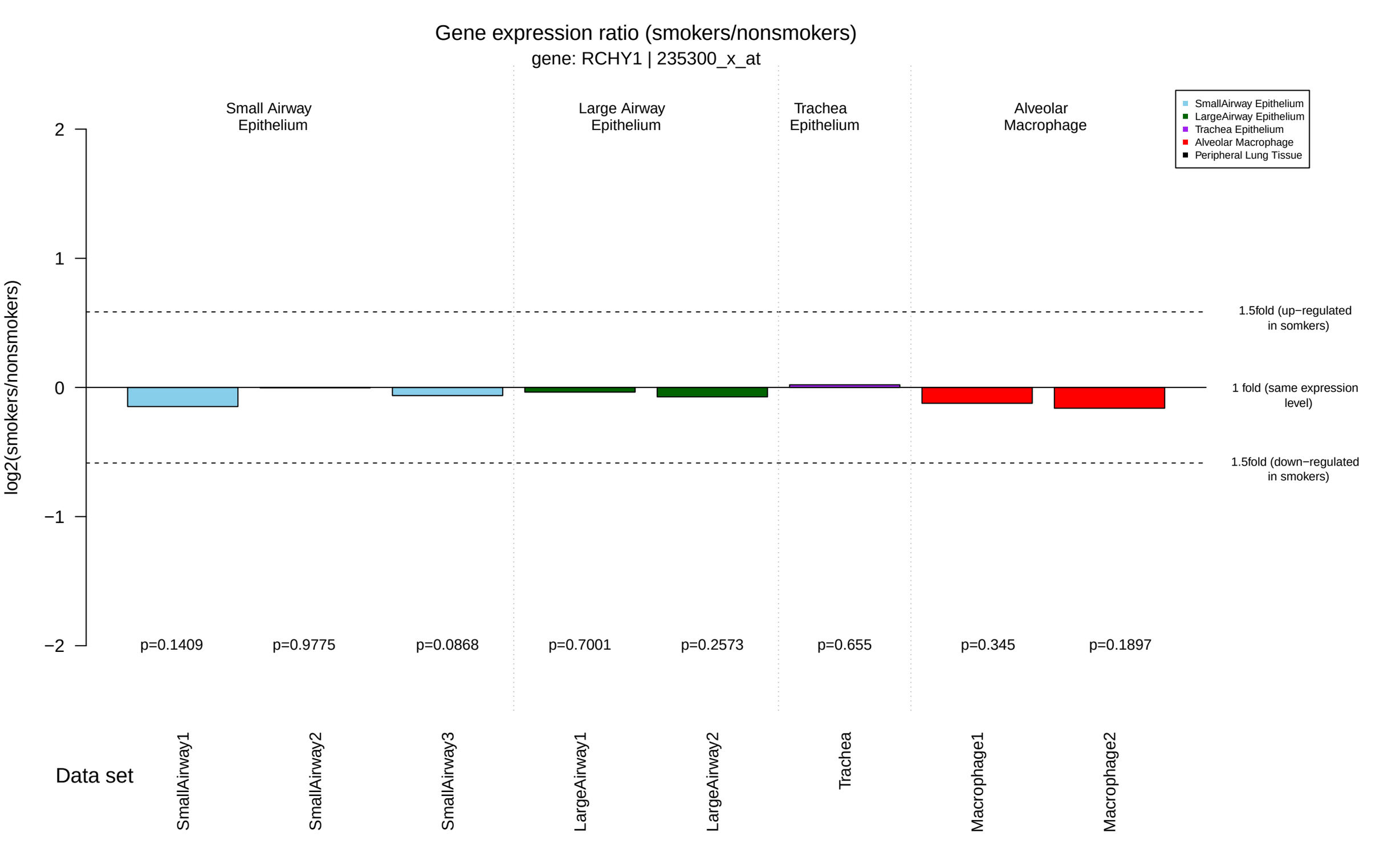
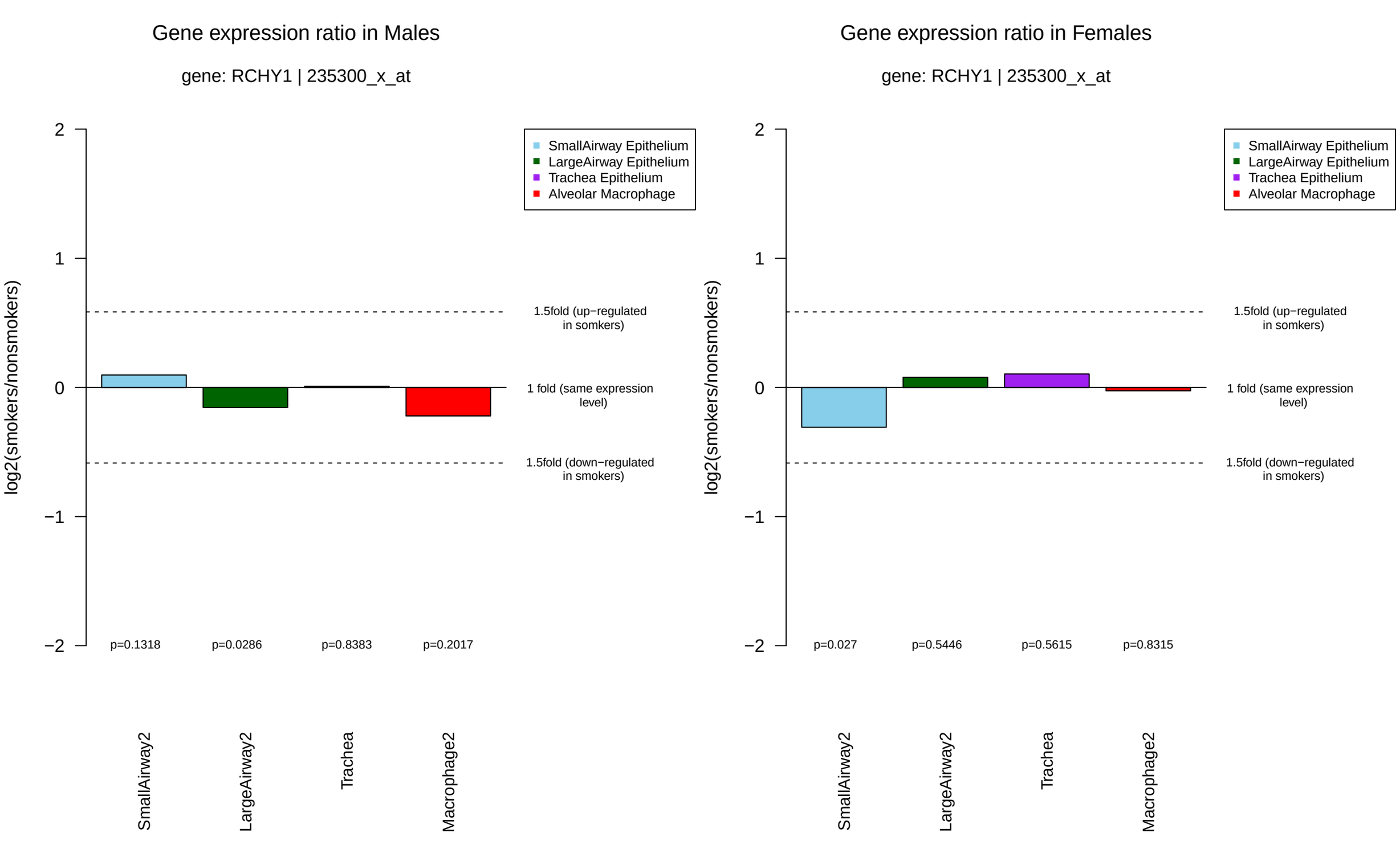
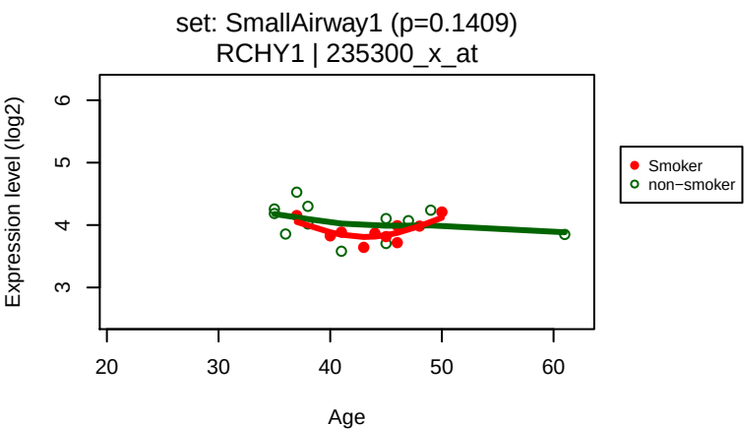
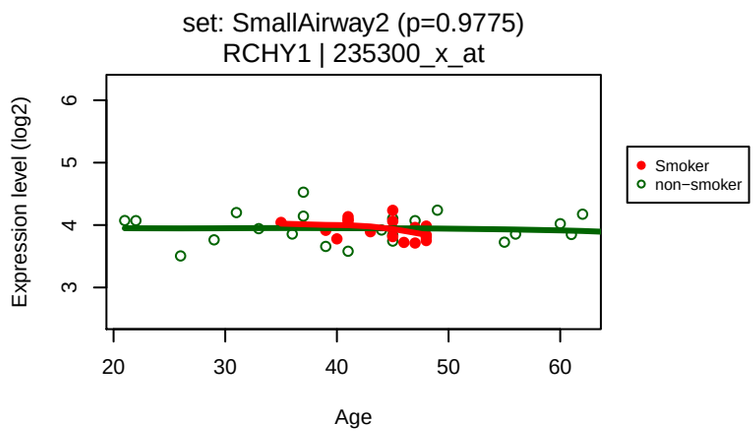
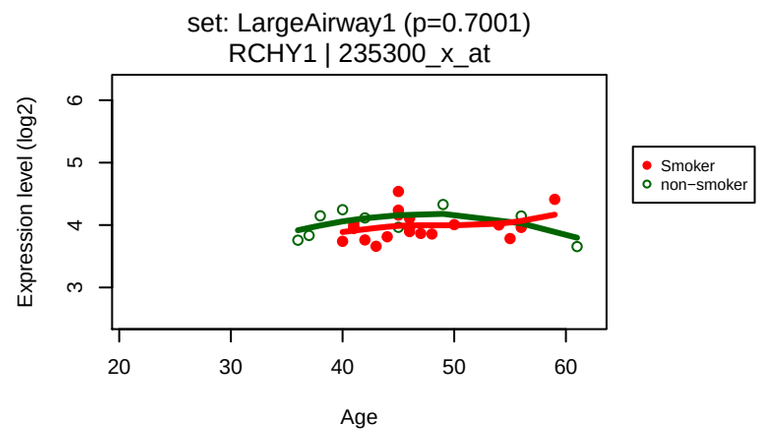

Gene expression of smokers and nonsmokers
gene: RCHY1 | 235300_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1409
p=0.9775
p=0.0868
p=0.7001
p=0.2573
p=0.655
p=0.345
p=0.1897
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

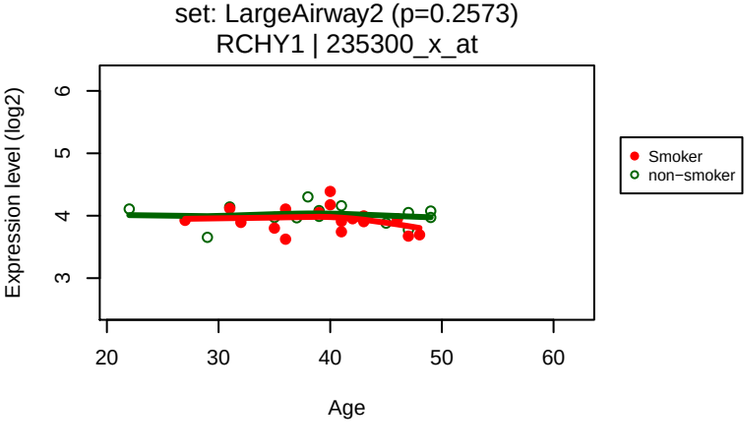

Gene expression ratio (smokers/nonsmokers)
gene: RCHY1 | 235300_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1409
p=0.9775
p=0.0868
p=0.7001
p=0.2573
p=0.655
p=0.345
p=0.1897
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

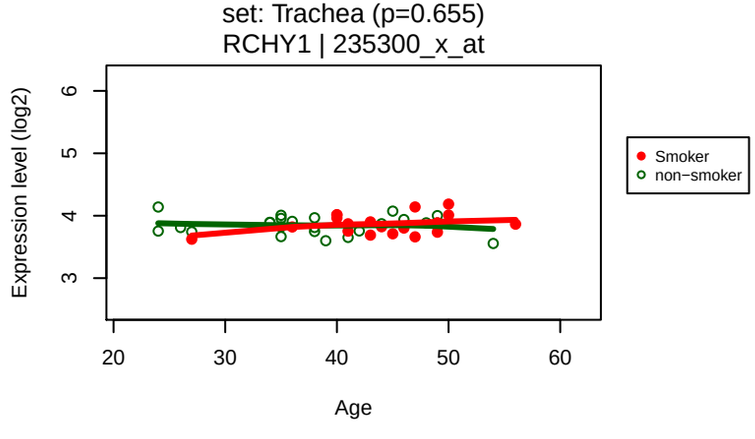

Gene expression ratio in Males
gene: RCHY1 | 235300_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1318
p=0.0286
p=0.8383
p=0.2017
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RCHY1 | 235300_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.027
p=0.5446
p=0.5615
p=0.8315
SmallAirway2
LargeAirway2
Trachea
Macrophage2

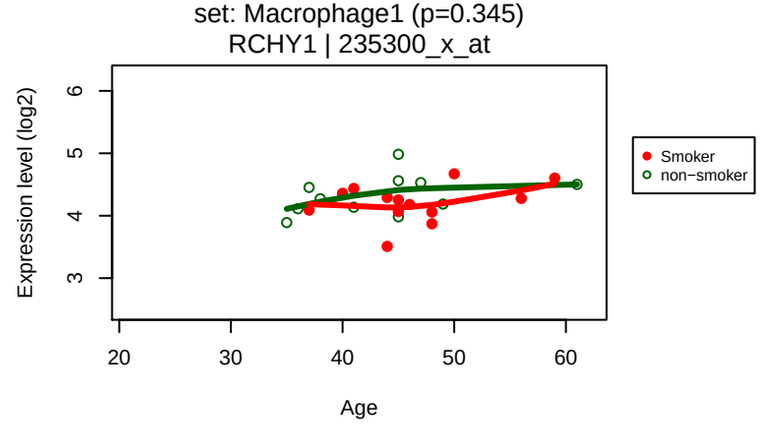

set: SmallAirway1 (p=0.1409)
RCHY1 | 235300_x_at
Expression level (log2)
Age
Smoker
non−smoker

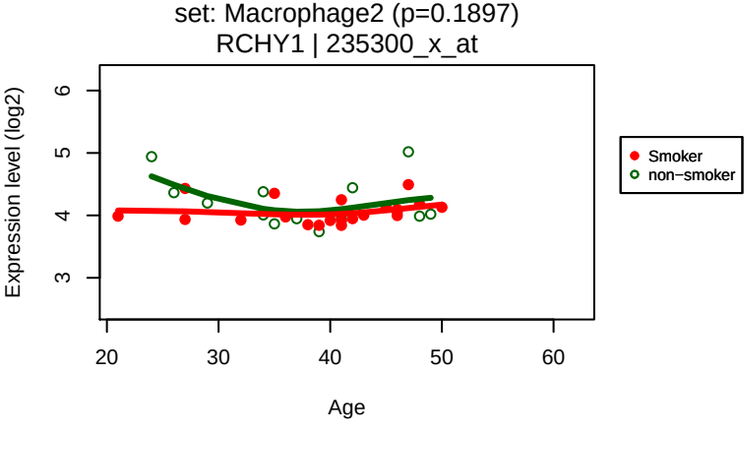

set: SmallAirway2 (p=0.9775)
RCHY1 | 235300_x_at
Expression level (log2)
Age
Smoker
non−smoker

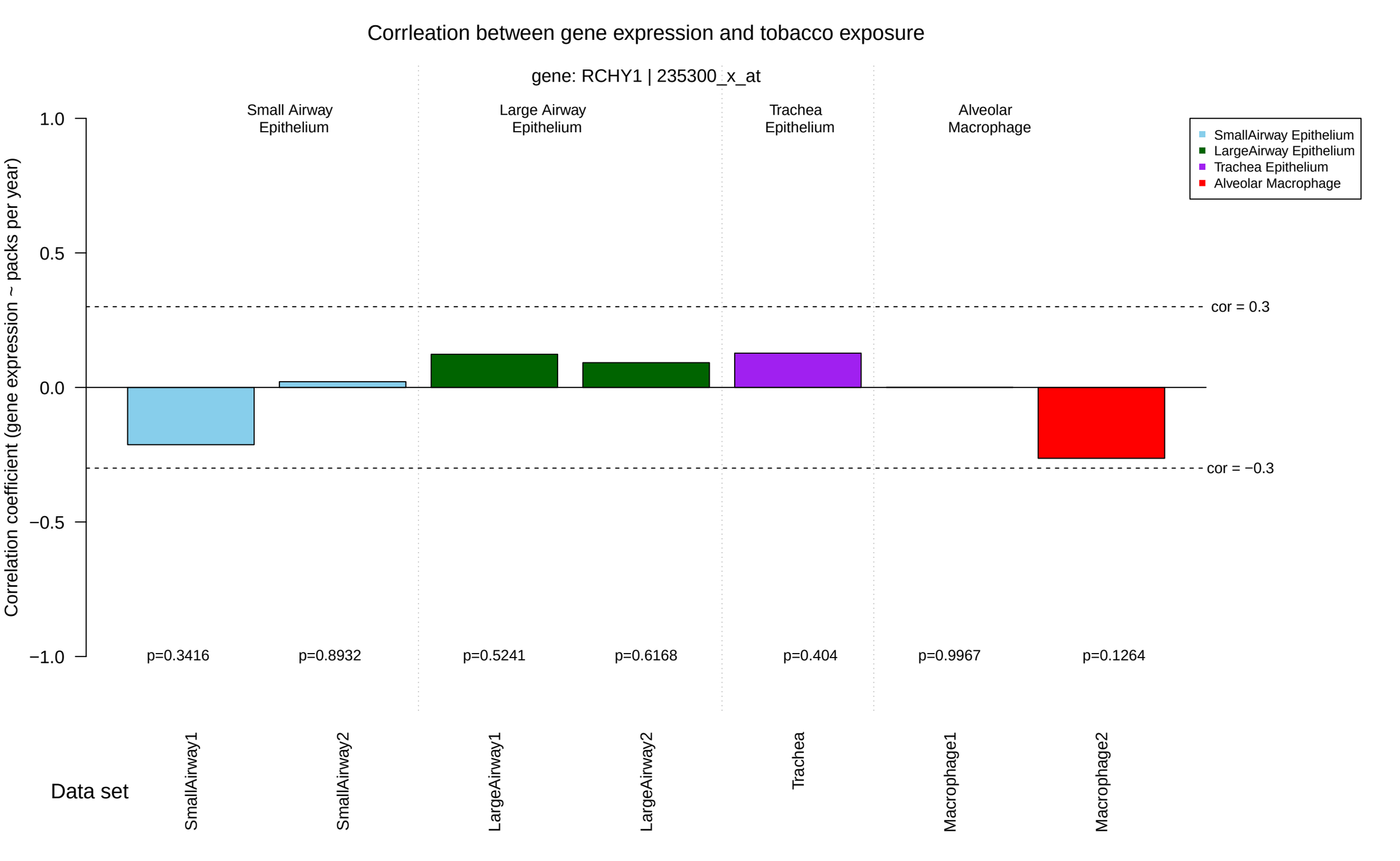

set: LargeAirway1 (p=0.7001)
RCHY1 | 235300_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2573)
RCHY1 | 235300_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.655)
RCHY1 | 235300_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.345)
RCHY1 | 235300_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1897)
RCHY1 | 235300_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RCHY1 | 235300_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3416
p=0.8932
p=0.5241
p=0.6168
p=0.404
p=0.9967
p=0.1264
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2